E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23656667

RECEIVED 09.05.2026

ACCEPTED 12.05.2026

PUBLISHED 30-06-2026

ENZYMES RESPONSIVE HYDROGELS FOR THE TREATMENT OF CANCER THERAPY

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ABSTRACT:

For the solution of this problem and improvement of anti-cancer therapy efficacy, innovative techniques are needed. The traditional approach to cancer treatment is characterized by poor selectivity, system toxicity, and side effects. Nowadays, one of the approaches to treatment of tumors is the development of a smart drug delivery system. Currently, there has been an intensive scientific interest in enzyme-sensitive hydrogels owing to their specific properties.

More precisely, enzyme-sensitive hydrogels represent three-dimensional polymer structures that experience certain chemical changes or decomposition under the number of particular enzymes abundant in a tumor environment. Among enzymes that act as triggers in this process, there can be found matrix metalloproteinases, and lipases that regulate the local drug release.

The current review paper covers classification and types of enzyme-responsive hydrogels, as well as the peculiarities of their work mechanism and synthesis techniques. Furthermore, modern trends connected with enzyme-responsive hydrogels are presented and . The problems with the practical usage of the mentioned hydrogels are analyzed as well.

Overall, it should be stated that enzyme-responsive hydrogels are a relatively new, but rather promising tool for treating cancers. Currently, among the most widespread diseases leading to mortality around the world, there is cancer.

Keywords: - *Enzyme-responsive hydrogels, Cancer therapy, Smart drug delivery, Tumor microenvironment, Controlled release, Biomaterials.*

1. Introduction:-

Cancer is one of those diseases that cannot be cured owing to many aspects associated with its pathogenesis. Despite the progress in diagnostic techniques and methods used in cancer treatment, the latter still remains a serious health problem worldwide. The traditional methods of cancer treatment involve chemotherapy, radiation, and surgery. Nonetheless, these strategies suffer from their high invasiveness and non-specificity that cause harm to adjacent healthy tissue structures. Anticancer agents are administered systemically that results in systemic toxicity, low efficacy of treatment, and development of drug resistance(Solanki, 2024).

As an innovative approach that would help in solving the issue of cancer therapy, targeted delivery of drugs came into use. By targeting the active substance at the tumor site, one may achieve greater therapeutic effect with lower side effects for a patient. There are numerous drug delivery systems; among all the available drug delivery strategies, one can single out hydrogels as a promising material(Jin et al., 2024).

A hydrogel is a three-dimensional, water-attracting polymer network capable of holding a substantial amount of water or biological fluid without dissolving. Due to the resemblance of hydrogels' architecture to that of the natural extracellular matrix, they have high biocompatibility and can be used for biomedical applications(Liu et al., 2025). Hydrogels can also be designed to provide controlled release of drugs, adjustable mechanical properties, and responsiveness to stimuli such as pH value, temperature, light, and enzymes.

The enzyme-triggered process stands out among the mentioned stimuli for its high specificity and frequent overexpression in pathological conditions like cancer. The tumor microenvironment reportedly possesses increased concentrations of specific enzymes, including matrix metalloproteinases (MMPs), proteases, and hyaluronidases, whose function is essential for the progression of tumors and metastasis. By taking advantage of this property, enzyme-responsive hydrogels include enzyme-sensitive linkages in their architecture, which are susceptible to cleavage by the corresponding enzymes.

Such enzymes can trigger hydrogel degradation or structural changes and, as a consequence, stimulate drug release. This enzyme-mediated process provides excellent control of drug delivery since drugs are released only at the targeted location, namely, within tumors. Therefore, enzyme-responsive hydrogels offer many benefits, such as increased efficiency, reduced side effects, and improved patient compliance(Yilmaz, 2025).

The objective of this review article is to summarize information regarding enzyme-sensitive hydrogels, highlighting aspects such as design concepts, mode of action, classification, methods of fabrication, and

application in cancer treatment. Recent advances and future prospects have also been elaborated upon in this evolving area(Liu et al., 2025).

2. Hydrogels: classification and properties

Amongst soft materials, hydrogels constitute an important category because of the unique nature of their crosslinked polymer networks that contain a substantial amount of water. Due to their special properties, hydrogels find numerous applications for drug delivery, tissue engineering, wound dressing, and regenerative medicines.

2.1 Classification of Hydrogels

Hydrogels can be categorized based on their source, cross-linking nature, and their response to external stimulus.

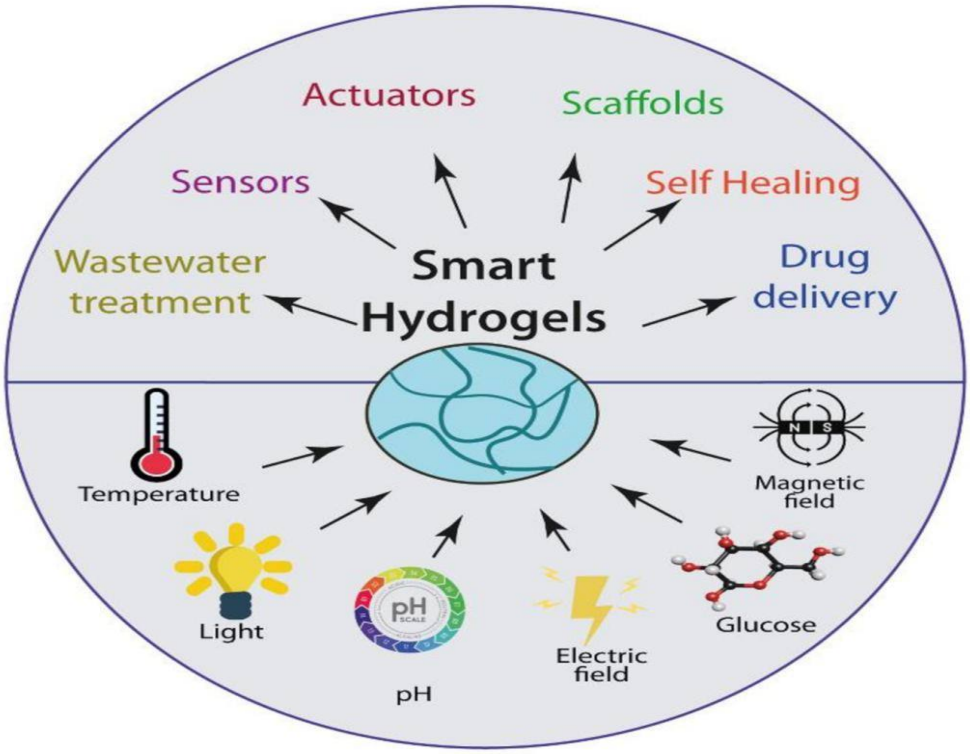


Fig. 1: *Classification of hydrogels based on source (natural and synthetic), crosslinking type (physical and chemical), and stimuli-responsiveness.*

(a) According to the source:

Natural hydrogels: Obtained from natural polymers like chitosan, alginate, gelatin, and hyaluronic acid. The natural hydrogels tend to be biocompatible and degradable; however, their strength tends to be low.

Synthetic hydrogels: Can be obtained from synthetic polymers like PEG, polyacrylamide, and polyvinyl alcohol. Synthetic hydrogels tend to provide better control over properties of interest, albeit at the expense of reducing biocompatibility.

(b) Depending upon cross-linking:

Physical hydrogels: Such gels form due to non-covalent bonding like hydrogen bonding, ionic bonding, or even hydrophobic bonding. In general, they are reversible and can undergo self-healing.

Chemical hydrogels: The cross-linking involves formation of covalent bonds between the chains of the polymers.

(c) Depending on responsiveness:

Stimuli-responsive hydrogels: Such hydrogels respond to variations in their environment, such as changes in pH, temperature, illumination, or enzyme presence. Enzyme-responsive hydrogels are included in this group and are highly valuable for targeted drug delivery applications(Pi et al., 2025).

2.2 Main Features of Hydrogels

The efficacy of hydrogels in biomedical applications depends greatly on their physicochemical features:

Absorption capacity: Hydrogels are capable of absorbing significant amounts of water, thus allowing effective diffusion and delivery of drugs.

Biocompatibility: Most hydrogels are non-toxic and biocompatible.

Biodegradability: Several types of hydrogels can decompose to non-toxic components, hence their applicability for in vivo use.

Mechanical stability: The mechanical stability of hydrogels can be adjusted according to their degree of cross-linking and chemical composition.

Porous structure: The porous structure of hydrogels makes them highly effective in terms of the loading and delivery of therapeutic substances.

Hydrogels are a perfect material choice for developing highly sophisticated drug delivery systems, especially those aimed at cancer treatment(Khurana et al., 2025).

Table 1: Comparison of Conventional vs Targeted Drug Delivery

Feature	Conventional Therapy	Targeted Hydrogel System
Specificity	Low	High
Side Effects	High	Reduced
Drug Release	Uncontrolled	Controlled
Efficiency	Moderate	High

3. Enzyme-responsive hydrogels: mechanism

Another type of hydrogels that is becoming popular today are enzyme-responsive hydrogels. They are defined as responsive biomaterials that change their structure or chemistry due to a specific enzyme activity. This class of hydrogels can be effectively used in cancer therapy because several enzymes are highly expressed in the tumor environment relative to healthy tissue(Omer et al., 2024).

3.1 The Tumor Microenvironment and Enzymatic Activity

As mentioned above, the tumor microenvironment is characterized by a highly complex system involving cancer cells, stroma, vascularization, and extracellular matrix(Li et al., 2023). It is known that the tumor microenvironment includes numerous enzymes that have significant effects on tumor progression(Zuo et al., 2026).

Some of the most important enzymes involved include:

- Matrix metalloproteinases (MMPs): Degradation of extracellular matrix.
- Proteases: For example, cathepsins are associated with protein degradation.
- Hyaluronidase: Degradation of hyaluronic acid in the extracellular matrix.

- Lipases and phospholipases: Involved in lipid metabolism and signaling in cancer cells.
- All enzymes mentioned are potential triggers for activating drug release in tumors.

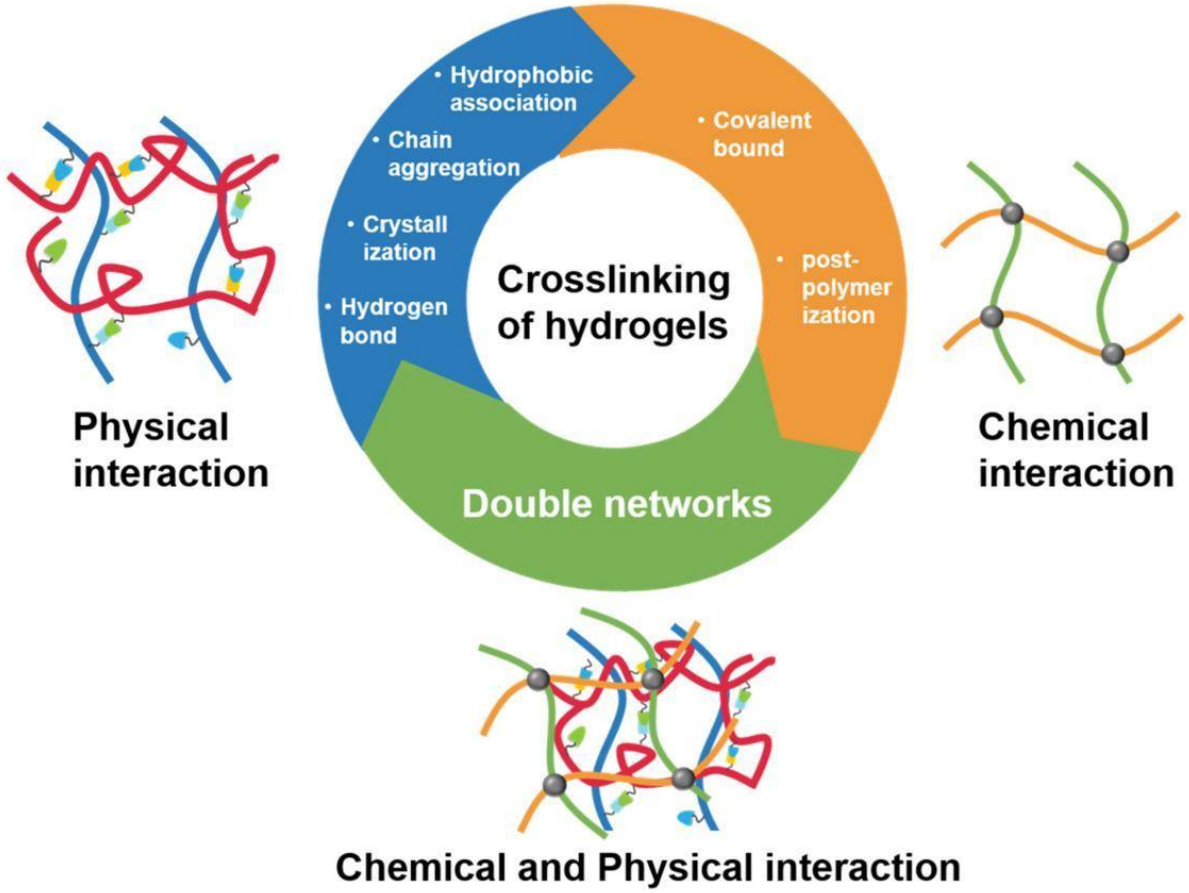


Fig. 2: *Tumor microenvironment illustrating the presence of enzymes such as matrix metalloproteinases (MMPs), proteases, lipases, and hyaluronidases involved in cancer progression.*

3.2 Drug Release Based on Enzyme Activity

The main idea behind the use of enzyme-responsive hydrogels is based on the presence of bonds sensitive to a specific enzyme activity in their structure. In other words, upon encountering a particular enzyme, the bonds are cleaved and thus alter the hydrogel structure(Sobczak, 2022).

Typically, it follows the steps below:

- Hydrogel implantation or delivery: The hydrogel with the drug inside is delivered into the body through injections or localized placement.

- Response to enzyme rich environment: The hydrogel is stable under physiological conditions but becomes active when the target enzymes are present.
- Cleavage of enzyme-sensitive bonds: Specific enzymes break certain peptide sequences or chemical links in the hydrogel matrix.
- Transformation: It causes disintegration of the hydrogel matrix.
- Sustained release of the drug: The drug is continuously released from the hydrogel matrix.

It ensures minimal loss of drug and reduced side effects(Tian et al., 2026).

3.3 Linkage Types That Enable Enzyme Sensitivity

Several linkages can be employed to create an enzyme-sensitivity response in the hydrogel matrix:

Peptide links: They can only be broken by specific proteolytic enzymes like the matrix metalloproteinases (MMPs).

- Ester linkage: It is sensitive to lipases and esterase enzymes.
- Amide bond: This linkage can be broken under enzymatic conditions.
- Polymeric backbone of polysaccharides: These can be degraded by specific enzymes such as hyaluronidase(Alshehri, 2024).

The kind of enzyme dictates the linkage used in the hydrogel.

3.4 Advantages of Enzyme Response Mechanism

- High degree of specificity due to enzyme sensitivity
- Continuous and controlled drug release
- Low off-target toxicity
- Enhanced efficiency in cancer therapy

Table 2: Classification of Hydrogels

Type	Example	Advantages	Limitations
Natural	Chitosan, Alginate	Biocompatible	Weak strength
Synthetic	PEG, PVA	Tunable properties	Less biodegradable

4. TYPES OF ENZYME-RESPONSIVE HYDROGELS

Hydrogels responding to particular enzymes can be classified according to their targets of interaction. Each type of enzyme-responsive hydrogel makes use of particular enzymes active in particular cancer microenvironments(Imran & Khan, 2025).

4.1 Protease-Responsive Hydrogels

Proteases are enzymes responsible for breaking down protein molecules into peptide or even amino acid residues. This group of enzymes is known for being up-regulated in many cancers and involved in tumor invasion and metastasis.

Protease-responsive hydrogels can be obtained with the help of particular peptide crosslinkers. After cleaving these crosslinkers with the enzymes, hydrogel structure gets destabilized and drug release begins.

Table 5: Types of Enzyme-Responsive Hydrogels

Type	Enzyme Target	Mechanism	Application
MMP-responsive	MMPs	Peptide cleavage	Tumor targeting
Protease-responsive	Proteases	Protein breakdown	Drug release
Lipase-responsive	Lipases	Ester cleavage	Hydrophobic drugs
HA-responsive	Hyaluronidase	HA degradation	ECM targeting

Applications:

- Targeted chemotherapy
- Local drug delivery to cancer tissues

4.2 MMP-Responsive Hydrogels

The family of matrix metalloproteinases is one of the best studied enzyme families involved in the hydrogel-based therapy of cancers. These enzymes actively degrade extracellular matrix proteins and are strongly up-regulated in several cancer types(Hoffman, 2012).

MMP-Responsive hydrogels utilize certain peptide sequences recognized by MMP enzymes and act as substrates for them. Once such hydrogels come into contact with an MMP-enriched microenvironment, they get degraded and the drug gets released(Kopeček, 2007).

Advantages:

- Specificity to tumor tissue
- Drug release at invasive tumor sites

4.3 Lipase-Responsive Hydrogels

Lipases are enzymes that facilitate lipid degradation by converting them to fatty acids and glycerols. Lipases are correlated with the metabolic processes of cancerous cells.

Lipase-responsive hydrogels contain ester bonds that are susceptible to enzymatic degradation by lipase enzymes. Such hydrogels can prove to be highly suitable for the controlled delivery of hydrophobic pharmaceuticals(Buwalda et al., 2017).

4.4 Hyaluronidase-Responsive Hydrogels

Hyaluronidase is an enzyme responsible for the breakdown of hyaluronic acid, which constitutes a part of the extracellular matrix(Caló & Khutoryanskiy, 2015). The enzyme is frequently overexpressed in cancers.

Hydrogels composed of hyaluronic acid may be designed to degrade upon encountering hyaluronidase; as such, they are efficient tumor-targeting platforms for drug delivery.

4.5 Multi-Enzyme Responsive Hydrogels

In order to achieve enhanced selectivity of hydrogels toward tumors, multi-enzyme responsive hydrogels have been created, featuring enzyme-sensitive linkages within one network.

Advantages:

- Increased specificity

- Improved regulation of drug delivery
- Ability to target tumor microenvironments effectively

5. Synthesis and fabrication methodsof enzyme-responsive hydrogels

It is very important to synthesize and prepare enzyme-responsive hydrogels because it allows to control the properties of the hydrogel, its stability, structure, and drug release mechanisms.

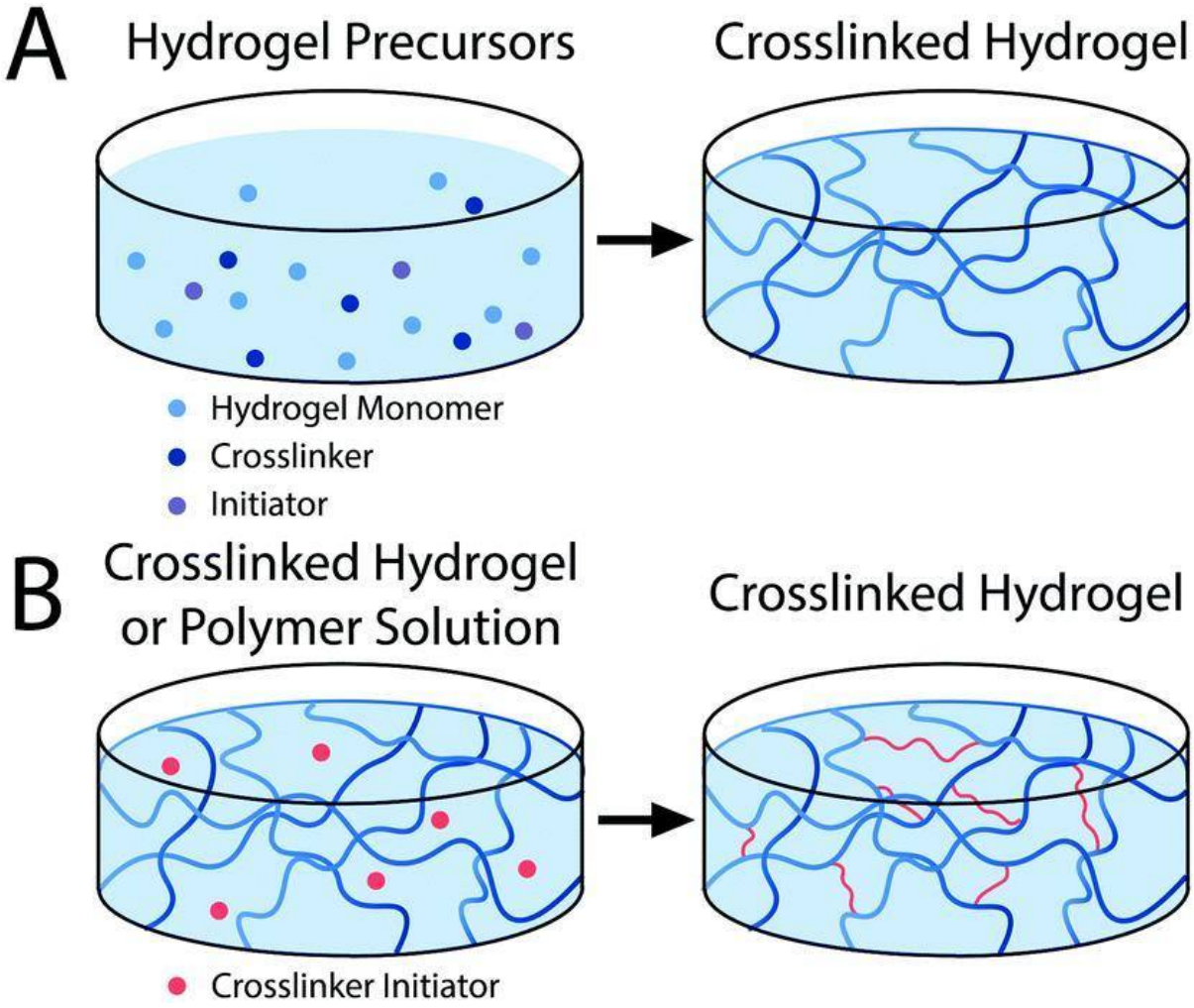


Fig. 3: *Schematic representation of synthesis and fabrication of enzyme-responsive hydrogels including polymer selection, crosslinking, and enzyme-sensitive linkages..*

5.1 Polymer Type Choice

Natural and synthetic polymers are used during hydrogel preparation. The examples of the natural polymer are chitosan, alginate, gelatin, hyaluronic acid, while the synthetic polymers include polyethylene glycol (PEG) and polyvinyl alcohol. In most cases, combined use of the two polymers allows receiving better characteristics of the final product.

5.2 Methods of Hydrogel Formation

During formation of hydrogel, polymer chains should be cross-linked. There are two types of cross-linking: chemical and physical. Physical cross-linking includes such non-covalent interactions as hydrogen bonding, ionic bonding, and van der Waals force; as a result, the hydrogel can be easily destroyed. Chemical cross-linking means that covalent bond is created between molecules

5.3 Enzyme-Sensitive Linkage

Enzyme-responsive hydrogels contain specific bonds sensitive to degradation: peptide, ester, polysaccharide backbones. These linkages can be cleaved under action of such enzymes as matrix metalloproteinases, proteases, hyaluronidase, etc.

5.4 Advanced Methods for Fabricating Hydrogels

Advanced methods such as click chemistry have made it possible to generate hydrogels rapidly and effectively under relatively mild conditions, enabling precise functional group modification. Self-assembly approaches are also popular, wherein molecules spontaneously self-assemble into an organized network structure without requiring harsh reagents.

5.5 Injectable Hydrogels

Injectable hydrogels can be delivered to targeted sites in liquid form and then converted to gel state upon reaching the targeted sites. They allow localized drug delivery, thus enhancing therapeutic efficacy and reducing unwanted side effects.

Table 6: Fabrication Techniques of Enzyme-Responsive Hydrogels

Method	Principle	Advantages	Limitations
Physical crosslinking	Non-covalent interactions	Reversible, simple	Low strength

Chemical crosslinking	Covalent bonding	Stable, durable	Toxic reagents possible
Click chemistry	Specific fast reactions	High precision	Costly
Self-assembly	Spontaneous organization	Mild conditions	Limited control

6. Applications of enzyme-responsive hydrogels in cancer therapy

The utilization of enzyme-responsive hydrogels is increasingly important in the area of cancer treatment, since they provide the means for achieving targeted delivery of drug molecules. Enzyme activity is usually higher within the vicinity of tumors compared to the rest of the body. Thus, enzyme-responsive hydrogels can be exploited to deliver therapeutic cargo at a targeted site.

6.1 Targeted Drug Delivery

Targeted delivery of the drug is one of the most popular applications of enzyme-responsive hydrogels. Such hydrogels are stable in normal tissues but degraded within the vicinity of tumor tissues due to enzymes. Thus, the drug is delivered at a target site, minimizing the potential adverse effects on healthy tissues(Peppas et al., 2000).

6.2 Controlled and Sustained Release

Controlled and sustained drug release is provided via enzyme-triggered degradation of hydrogels. The process of degradation allows controlling the duration of treatment. Also, the speed of release can be tailored according to specific requirements(Li & Mooney, 2016).

6.3 Injectable Hydrogels

Injectable hydrogels enable minimally invasive delivery of drugs. These hydrogels are capable of forming gels in situ and providing localized release of therapeutic agents.

6.4 Combination Therapy

Enzyme-responsive hydrogels can be used in combination therapies, such as chemotherapy combined with immunotherapy or photothermal therapy.

Table 9: Applications in Cancer Therapy

Application	Mechanism	Benefit
Targeted delivery	Enzyme-triggered release	High specificity
Sustained release	Gradual degradation	Long-term effect
Injectable systems	Localized gel formation	Reduced toxicity
Combination therapy	Multi-drug delivery	Enhanced efficacy

7. Advantages and limitations

Advantages of Enzyme-responsive Hydrogels over Drug Delivery Methods

The specific targeting of enzyme-responsive hydrogels towards cancer enzymes is one of the key advantages over traditional methods of drug administration. In addition to that, biodegradability makes these hydrogels non-toxic and eliminates possible harm caused by drug delivery agents. Moreover, they can perform multiple functions(Webber & Langer, 2017).

Limitations of Enzyme-responsive Hydrogels

However, there are some disadvantages associated with these materials. Firstly, physical stability might be an issue because hydrogels could be affected by environmental factors. Designing such systems requires additional expenses. Lastly, the different activity of enzymes depending on patients should be taken into account.

Table 10: Advantages vs Limitations

Advantages	Limitations
Target specificity	Stability issues
Controlled release	Complex design
Reduced toxicity	High cost
Biodegradable	Clinical variability

8. Recent advances

Modern developments in enzyme-responsive hydrogels include increasing their efficacy and usefulness in the clinic. For example, the combination of nanoparticles in a polymer network of hydrogels leads to increased loading ability and higher efficacy of therapy. Another promising development is related to multi-stimuli hydrogels that can respond not only to enzymes but also to changes in other conditions like pH or temperature in order to control the drug release(Yu et al., 2018).

Moreover, because of the latest advancements in bioengineering and nanotechnology, personalized hydrogel systems according to patients' attributes can now be engineered. The utilization of such hydrogel systems in cancer treatment provides great potential in the future.

9. Challenges and future prospects

Though enzyme-responsive hydrogels exhibit tremendous success, several challenges exist in their design and application for cancer treatment. The most prominent challenge is the scaling of these systems from the laboratory to clinical trials. Among the concerns that must be addressed include scaling, reproducibility, and regulatory considerations.

In the next few years, research will be focused on improving the stability and scalability of enzyme-responsive hydrogels. Systems that respond to multiple stimuli with the integration of modern technological concepts like artificial intelligence and precision medicine could serve as future directions. As more research is conducted, enzyme-responsive hydrogels may emerge as a vital tool for cancer treatment in the future.

10. Conclusion

The application of enzyme-responsive hydrogels is one of the new promising approaches in the area of cancer therapy due to their capacity to react only when coming into contact with the enzymes released by the tumor tissue, which makes such a technique much more effective and ensures less adverse effects on healthy tissues.

There are a variety of drugs that can be used within the enzyme-responsive hydrogels. However, the issue of stability and scalability should be solved before using these hydrogels in clinical practice.

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